

Pipeline Dataset Brief

File: PIPELINE_CAA_00_COMPLETE.csv

Dimensions: 66,655 rows, 35 columns

Source: EPA ECHO — CAA Compliance Pipeline

Documentation: <https://echo.epa.gov/tools/data-downloads/caa-pipeline-download-summary>

1 Unit of Observation

Each row is a **violation** $\times$ **enforcement action** $\times$ **evaluation** combination. A single violation can appear in multiple rows if it is linked to multiple enforcement actions or evaluations. The unique key is SOURCE_ID + VIOL_ACTIVITY_ID + EA_ACTIVITY_ID + EVAL_ACTIVITY_ID.

- 52,346 distinct violations (VIOL_ACTIVITY_ID)
- 20,251 distinct facilities (SOURCE_ID, equivalent to PGM_SYS_ID in ICIS-Air)
- 78% of violations appear in exactly 1 row; 16% in 2 rows; 4% in 3 rows; <1% in 4+

2 Data Generating Process

According to EPA documentation, the ECHO pipeline “shows links between Compliance Monitoring Activities (CMA) to any related violations and/or enforcement actions.” FOUND_VIOLATION = Y for all 66,655 rows. However, 10.8% of rows carry system-generated VIOL_ACTIVITY_IDs rather than real violations. According to the EPA, these IDs “did not have an actual violation activity identification number” and were “system generated for purposes of creating the pipeline table.”

PIPELINE_FLAG

According to EPA documentation: “An internally generated flag to indicate at least [one] violation (VIOL_ACTIVITY_ID) linked to an evaluation and/or enforcement action.” When set to Y, the pipeline table is displayed on the facility’s ECHO Detailed Facility Report.

Flag	Rows	Facilities	Has EA	Has Eval
Y	58,251 (87.5%)	15,792	78%	52.4%
N	8,404 (12.5%)	4,459	0%	0%

Empirical pattern: All 8,404 rows with PIPELINE_FLAG = N have EA_FLAG = N and EVAL_FLAG = N with no linked enforcement actions or evaluations. These rows have violation-level fields (type, dates). It’s not super clear from the documentation whether N rows once had linkages that were removed or if they never had linkages.

Placeholder rows (no actual violation)

Not every row represents a real violation. 7,193 rows (10.8%) carry **system-generated** VIOL_ACTIVITY_IDS. The EPA documentation states that values starting with 9906 or 9913 “did not have an actual violation activity identification number” and were “system generated for purposes of creating the pipeline table.” These fall into two categories:

1. VIOL_TYPE = NA (4,067 rows, IDs starting with 9906): These rows have EA_FLAG = Y but no violation dates or type.
2. VIOL_TYPE = ‘Linked to Viol. Below’ (3,126 rows, IDs starting with 9913): These rows have EA_FLAG = N, EVAL_FLAG = N, and no violation dates.

The remaining 59,462 rows (89.2%) carry real violation IDs (10-digit, starting with 3400 or 3600) and a substantive VIOL_TYPE of HPV or FRV.

“Date Out of Range” enforcement actions

EA_TYPE = ‘Date Out of Range’ appears in 1,116 rows (2.5% of EA-linked rows). All 1,116 have no EA_DATE recorded. The cause is not documented by EPA.

3 Summary Statistics

Violations

Filtering to real violations ($\text{VIOL_TYPE} \in \{\text{HPV}, \text{FRV}\}$), the pipeline contains 45,164 distinct violations across 59,462 rows. FRV violations outnumber HPV roughly 2.4:1 (42,037 rows vs. 17,425 rows). Of the 45,164 distinct real violations, 81.4% have PIPELINE_FLAG = Y and 18.6% have PIPELINE_FLAG = N.

Among violations with valid dates ($n = 9,222$), the median time to resolution is **180 days** (mean 304 days; IQR 56–384 days).

Enforcement actions

68% of rows (45,340) have an enforcement action linked ($\text{EA_FLAG} = \text{Y}$). The distribution of EA types:

EA Type	Rows	% of EA rows
Notice of Violation	23,818	52.5%
Administrative - Formal	16,782	37.0%
Warning Letter	2,318	5.1%
Date Out of Range	1,116	2.5%
Judicial	380	0.8%
Other / State-local	926	2.0%

Penalties and compliance action costs

14,427 rows carry a non-zero penalty (EA_PENALTY_AMT), totaling \$1.93 billion. The distribution is heavily right-skewed: median \$7,350, mean \$133,818, max \$47.8M.

Compliance action costs ($\text{EA_COMP_ACTION_COST}$) are sparse, only 94 rows have a non-zero value, but large in magnitude, totaling \$849M.

Lead agency

VIOL_LEAD_AGENCY identifies which agency leads the violation case.

Agency	Rows	Facilities	% with EA	Total Penalties	Median Penalty
State	59,262 (88.9%)	18,812	69.5%	\$1.61B	\$6,750
EPA	194 (0.3%)	114	74.7%	\$3.5M	\$75,000
Local	6 (<0.1%)	4	100%	\$964K	\$482K

The remaining 7,193 rows (10.8%) have no lead agency—these are the placeholder rows with system-generated VIOL_ACTIVITY_IDs. State agencies lead the vast majority of cases. EPA-led cases are rare but carry substantially higher median penalties (\$75,000 vs. \$6,750 for state-led).

4 Example Rows

Example A — Unresolved FRV, no enforcement action yet

Field	Value
SOURCE_ID	OK0000004005100121
AIR_NAME	DCP Operating Co LP / Wild Hog Booster Station
PIPELINE_FLAG	Y
EVAL_FLAG / EVAL_TYPE_DESC	Y / Self-Disclosure
VIOL_TYPE	FRV
VIOL_START_DATE	04/18/2024
VIOL_END_DATE	—
EA_FLAG	N
EA_PENALTY_AMT	—

A facility self-disclosed a federally reportable violation. No enforcement action has been taken. Violation is still in the pipeline.

Example B — HPV discovered by FCE, formal action with penalty

Field	Value
SOURCE_ID	NVCCH00032003A0019
AIR_NAME	Titanium Metals Corp. Henderson Facility
PIPELINE_FLAG	Y
EVAL_FLAG / EVAL_TYPE_DESC	Y / FCE On-Site
EVAL_DATE	05/13/2024
VIOL_TYPE	HPV
VIOL_START_DATE	08/05/2024
EA_FLAG / EA_TYPE	Y / Administrative - Formal
EA_DATE	07/24/2024
EA_PENALTY_AMT	\$11,150

An on-site FCE is linked to an HPV with a formal administrative action and an \$11,150 penalty. PIPELINE_FLAG = Y despite the penalty being assessed.

Example C — One violation, three rows (multiple EAs)

SOURCE_ID	VIOL_ACTIVITY_ID	EA_TYPE	EA_DATE
MS0000002812100033	3400330404	Notice of Violation	10/09/2001
MS0000002812100033	3400330404	Date Out of Range	—
MS0000002812100033	3400330404	Date Out of Range	—

A single HPV at Holmes Group / Rival Division is linked to three enforcement action records. Two have EA_TYPE = “Date Out of Range” with no EA_DATE. The violation appears in three rows because each EA linkage generates a separate row.

Example D — Violation with Pipeline 'N'

Field	Value
SOURCE_ID	NC0000003703200251
AIR_NAME	North Carolina Central University
PIPELINE_FLAG	N
VIOL_TYPE	FRV
VIOL_START_DATE	11/18/2021
VIOL_END_DATE	05/17/2023
EA_FLAG	N
EVAL_FLAG	N

PIPELINE_FLAG = N with a violation start and end date spanning ~18 months. Consistent with the pattern described above: EA_FLAG = N and EVAL_FLAG = N.

5 Key Variables

Violation fields:

- VIOL_TYPE: HPV (high priority, 26%) or FRV (federally reportable, 63%). The remaining 11% are placeholder rows: 6% NA (system-generated IDs with no actual violation) and 5% “Linked to Viol. Below.”
- VIOL_ACTIVITY_ID: the key linking violations to CMAs and enforcement actions. IDs starting with 9906 or 9913 are system-generated placeholders (per EPA documentation).
- VIOL_START_DATE: when the violation was determined; missing for 10.8% of rows (all placeholder rows lack dates).
- VIOL_END_DATE: violation end date; missing for 74.6% of rows.
- VIOL_PROGRAMS: which CAA programs were violated (CAATVP, CAASIP, CAAMACT, etc.).

Enforcement action fields (only for EA_FLAG = Y, 68% of rows):

- EA_TYPE: Notice of Violation (52%), Administrative - Formal (37%), Warning Letter (5%), Date Out of Range (2.5%), Judicial (1%).

- **EA_DATE**: date of the enforcement action. Missing for all “Date Out of Range” rows.
 - **EA_PENALTY_AMT**: monetary penalty; non-zero for 14,427 rows. Median \$7,350, mean \$133,818, max \$47.8M.
 - **EA_COMP_ACTION_COST**: supplemental environmental project cost; non-zero for only 94 rows.
- Evaluation fields** (only for **EVAL_FLAG** = Y, 46% of rows):
- **EVAL_TYPE_DESC**: PCE Off-Site (26%), FCE On-Site (15%), Investigation (11%), Self-Disclosure (11%).
 - **EVAL_LEAD_AGENCY**: State (84%), EPA (8%), Local (8%).
 - **EVAL_ACTIVITY_ID**: set to -9999 when no evaluation is linked (43,084 rows, 65%).
- Flags and metadata:**
- **PIPELINE_FLAG**: per EPA, “an internally generated flag to indicate at least [one] violation linked to an evaluation and/or enforcement action.” In the data, all N rows have **EA_FLAG** = N and **EVAL_FLAG** = N.
 - **EVAL_FLAG**: per EPA, “an internally generated flag to indicate if there is at least one Compliance Monitoring Activity (CMA) (evaluation) to the violation activity id.”
 - **EA_FLAG**: per EPA, “a system generated flag indicating if there is an enforcement action.”
 - **VIOL_FLAG**: per EPA, “a system generated flag indicating a violation is present. Entries noted as violation not found will have a generated **VIOL_ACTIVITY_ID** to use as links between Evaluations and Enforcement Actions.”
 - **OFFICIAL_FLAG**: per EPA, “this flag is set to ‘Y’ when an entry is counted as compliance monitoring strategy activities.”
 - **FEA_ISSUE_DATE_FLAG**: Y = a formal enforcement action issue date exists (224 rows).
 - **SORT_ORDER**, **SORT_DATE**: system-generated fields to aid linkage creation and display ordering.

6 Join Keys to Other Datasets

Pipeline field	ICIS-Air equivalent	Join rate
SOURCE_ID	PGM_SYS_ID	100% (20,251/20,251)
REGISTRY_ID	REGISTRY_ID (FRS)	99.9% (19,672/19,687)
VIOL_ACTIVITY_ID	ACTIVITY_ID (VIOLATION_HISTORY)	86.3% (45,164/52,346)
EA_ACTIVITY_ID	ACTIVITY_ID (FORMAL/INFORMAL)	98.9% (40,024/40,478)
EVAL_ACTIVITY_ID	ACTIVITY_ID (FCES_PCES)	66.4% (12,199/18,360)

7,182 pipeline violations are *not* in the ICIS-Air violation history table—these are the system-generated placeholder IDs. Conversely, 55,983 violation history records are *not* in the pipeline. The pipeline is not a superset of all violations.

7 What the Pipeline Adds Over Individual ICIS-Air Tables

The individual ICIS-Air tables share **ACTIVITY_ID** and **PGM_SYS_ID** but do not directly link violations to enforcement actions or evaluations. The pipeline pre-joins these linkages. Unique to the pipeline:

1. **Violation-to-enforcement linkage** — which EA is linked to which violation
2. **Violation resolution status** — **PIPELINE_FLAG** and **VIOL_END_DATE**
3. **Time-to-resolution** — from **VIOL_START_DATE** to **VIOL_END_DATE**